STAT 252 — Chi-Square Experiments

Guided Analysis Workbook

Cal Poly San Luis Obispo

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How to Use This Workbook

This workbook covers four experiments organized around two chi-square techniques:

- **Goodness of Fit** (Experiments 4 & 5): Does the observed distribution of a *single* categorical variable match a hypothesized distribution?
- **Test of Independence** (Experiments 6 & 7): Are *two* categorical variables associated, or are they independent?

For each experiment we follow the **SSVA framework**:

1. **State the research question.**
2. **Look at the raw data** in the Google Sheet.
3. **Examine summary statistics** (observed vs. expected counts).
4. **Interpret the visualization.**
5. **Run the analysis** (chi-square test statistic, p-value).
6. **Answer SSVA guiding questions.**
7. **Answer independent conceptual questions.**

Session code: section-8
Class Google Sheet: open in new tab
Report generated: April 15, 2026 at 11:46 AM

Tip

Before each experiment: open the class Google Sheet linked above and click the tab for that experiment (roulette, birthmonth, coffee, or pressure).

1. Experiment 4 — Goodness of Fit: Is the Roulette Wheel Rigged?

1.1 Research Question

> **When pooling all students' roulette spins, does the distribution of outcomes (pockets 1–6) match what we'd expect from a fair wheel?**

1.2 Step 1 — Look at the Raw Data

> Note
>
> **Open the class Google Sheet and click the roulette tab.** Each row is one student. The wheel_type column shows whether they had a fair or rigged wheel (they didn't know during the experiment). The pocket_1 through pocket_6 columns show how many times each pocket appeared in their 30 spins. Scan a few rows — do you notice any students with suspiciously many 6s?

First few students' roulette spin tallies

1s	2s	3s	4s	5s	6s
4	2	2	1	3	3
4	2	2	1	3	3
4	2	2	1	3	3

1.3 Step 2 — Summary Statistics

Observed vs. expected counts across all students' spins

Pocket	Observed count	Expected count (if fair)
1		
2		
3		
4		
5		
6		

1.4 Step 3 — Visualization

Describe Visualization

1.5 Step 4 — Analysis

Chi-square goodness-of-fit test

Quantity	Value
Chi-square statistic (χ^2)	
Degrees of freedom	
p-value	
Total spins pooled	

1.6 SSVA Guiding Questions — Experiment 4

Q1. What single categorical variable are we studying? What are its possible values?

Your answer:

Q2. What is the hypothesized (expected) distribution under the null?

Your answer:

Q3. Compare the observed and expected bars in the chart. Which face(s) depart most from expectation? What might explain this?

Your answer:

Q4. How does the chi-square statistic work? What would a value of 0 mean? What does a large value mean?

Your answer:

Q5. Based on the visualization and summary statistics alone, what is your initial conclusion?

Your answer:

1.7 Independent Conceptual Questions — Experiment 4

C1. What is the statistical question?

C2. What is the population? What is the sample?

C3. What is the observational unit?

C4. How many variables are in this analysis? Name them.

C5. State the null and alternative hypotheses.

C6. How are degrees of freedom calculated for this test?

C7. What potential confounding factors or design limitations exist?

C8. Based on the test statistic and p-value, what do you conclude?

2. Experiment 5 — Goodness of Fit: Birth Month Distribution

2.1 Research Question

Are birth months in our class uniformly distributed across the 12 months of the year?

2.2 Step 1 — Look at the Raw Data

Note

Open the class Google Sheet and click the birthmonth tab. Each row is one student with their birth month. How many total students submitted? Can you eyeball which months are most and least common?

First few rows of the birth month data

Birth Month
Feb
Apr
Mar

2.3 Step 2 — Summary Statistics

Observed vs. expected birth month counts

Month	Observed	Expected (uniform)
Jan		0.2
Feb		0.2
Mar		0.2
Apr		0.2
May		0.2
Jun		0.2
Jul		0.2
Aug		0.2
Sep		0.2
Oct		0.2
Nov		0.2
Dec		0.2

2.4 Step 3 — Visualization

Describe Visualization

2.5 Step 4 — Analysis

Chi-square goodness-of-fit test

Quantity	Value
Chi-square statistic (χ^2)	
Degrees of freedom	
p-value	
Total students	

2.6 SSVA Guiding Questions — Experiment 5

Q1. What single categorical variable are we studying? How many categories does it have?

Your answer:

Q2. What is the expected distribution under the null hypothesis?

Your answer:

Q3. Look at the bar chart. Which months, if any, stand out as having more or fewer students than expected?

Your answer:

Q4. Why might the uniform distribution actually be a poor model for real birth months?

Your answer:

Q5. Based on SSVA alone, does the distribution look uniform?

Your answer:

2.7 Independent Conceptual Questions — Experiment 5

C1. What is the statistical question?

C2. What is the population? What is the sample?

C3. What is the observational unit?

C4. State the null and alternative hypotheses.

C5. What are the degrees of freedom, and how are they calculated?

C6. What is a potential concern with expected cell counts here? Does it apply?

C7. Is this an experiment or an observational study?

C8. Based on the test statistic and p-value, what do you conclude?

3. Experiment 6 — Test of Independence: Coffee Drink & Where You're From

3.1 Research Question

> **Is the kind of coffee drink students prefer associated with whether they grew up on the coast or inland?**

> *No data found for the 'coffee' experiment in session `section-8`. Skipping.*

3.2 Step 1 — Look at the Raw Data

> Note
>
> **Open the class Google Sheet and click the coffee tab.** Each row records one student's hometown type (Coast or Inland) and their go-to coffee drink. Browse — do you see any pattern?

3.3 Step 2 — Summary Statistics

3.4 Step 3 — Visualization

Describe Visualization

3.5 Step 4 — Analysis

Quantity	Value
Chi-square statistic (χ^2)	
Degrees of freedom	
p-value	
Total students	

3.6 SSVA Guiding Questions — Experiment 6

Q1. What two categorical variables are we analyzing, and what are the dimensions of the table?

Your answer:

Q2. If origin and drink were independent, what would the bar chart look like?

Your answer:

Q3. Which cells depart most from expected counts? What does that tell you?

Your answer:

3.7 Independent Conceptual Questions — Experiment 6

C1. State the hypotheses.

C2. Degrees of freedom?

C3. Experiment or observational study? Can we conclude causation?

C4. Based on the test statistic and p-value, what do you conclude?

4. Experiment 7 — Test of Independence: Pressure Test (Stress & Performance)

4.1 Research Question

> **Does students' self-reported stress level predict how well they actually perform on a fast-paced reaction game?**

> *No data found for the 'pressure' experiment in session `section-8`. Skipping.*

4.2 How Performance Categories Are Assigned

After the 30-second reaction game, each student's **score** (number of correct taps) and **accuracy** (correct ÷ total attempts) are combined into a single performance category using the following rubric:

Category	Score threshold	Accuracy threshold	Plain-English meaning
High	≥ 16 correct	≥ 70%	Fast *and* accurate — both conditions must be true
Medium	≥ 8 correct	≥ 55%	Reasonably fast and reasonably accurate — both conditions must be true
Low	Below Medium thresholds	—	Missed at least one of the Medium conditions

Tip

Why two conditions? Speed alone can be gamed by tapping randomly; accuracy alone can be faked by tapping slowly. Requiring *both* score and accuracy to clear the bar ensures the category reflects genuine performance under pressure.

4.3 Step 1 — Look at the Raw Data

Note

Open the class Google Sheet and click the pressure tab. Each row records one student's self-reported stress level *before* the game and their measured performance category *during* the game (plus their raw score, accuracy, and average reaction time).

4.4 Step 2 — Summary Statistics

4.5 Step 3 — Visualization

Describe Visualization

4.6 Step 4 — Analysis

Quantity	Value
Chi-square statistic (χ^2)	
Degrees of freedom	
p-value	
Total students	

4.7 SSVA Guiding Questions — Experiment 7

Q1. What's special about how the two variables in this experiment were measured?

Your answer:

Q2. Dimensions of the contingency table and degrees of freedom?

Your answer:

Q3. If stress and performance were independent, what would the visualization show?

Your answer:

Q4. What story would you expect intuitively? Does the data agree?

Your answer:

4.8 Independent Conceptual Questions — Experiment 7

C1. State the hypotheses.

C2. Is this an experiment or an observational study? Justify carefully — there's a subtlety.

C3. What confounders could explain any association we find?

C4. Based on the test, what do you conclude?